

Working paper to be sent to Biological Reviews. *This journal accepts reviews that go a step beyond and incorporate a new perspective instead of just reviewing a given topic.*

Title 1: Moving towards an integrative approach for understanding biodiversity and ecosystem dynamics in a rapid changing world

Title 2: The BAD COMETA: moving towards an integrative approach for understanding biodiversity and ecosystem dynamics in a rapid changing world

Title 3: Biodiversity Multitrait Dynamics in Rapidly Changing Landscapes

Outline

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3. Moving towards a multi-trait-based approach for coevolutionary interactions in metaecosystems

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3.1.1. Why a trait-based approach?

Box 1.

A coevolutionary IBM to study biodiversity dynamics and ecosystem functioning in a rapid changing world.

3.2. The empirical data that will allow model testing

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1. General Introduction (Cecilia and Julia).

(Here we will set the arguments for the importance of a multi-trait coevolutionary approach to investigate biodiversity dynamics and the role of different rescue mechanisms in allowing species persistence in rapid changing landscapes)

(1) The biodiversity crisis in a rapid changing world: how species may respond to the consequences of anthropogenic threats -i.e. mutualistic/antagonistic partners loss, habitat loss and conversion- and persist, and how species persistence ability may affect ecosystem functioning.

Current anthropogenic threats to biodiversity and their shared consequences on ecosystem functioning. Land conversion, climate change, species invasions. Shared consequences on

biodiversity (alfa and beta species diversity), functional diversity (alfa and beta trait diversity). Impact on ecosystem functioning.

The potential of rescue mechanisms to prevent species, interactions and trait diversity loss in rapid changing landscapes. How ecological (rewiring, dispersal), evolutionary (adaptation, plasticity) and co-evolutionary processes that allow species persistence (i.e. rescue mechanisms) feedback among them and drive biodiversity dynamics. Consequences on ecosystem functioning.

(2) Among the evolutionary and coevolutionary processes, rapid evolutionary change, coevolution and niche construction may allow species to respond to anthropogenic threats.

Why rapid evolution and eco-evolutionary feedbacks are important in the context of anthropogenic threats. Long-term evolution (i.e. diversification) may not represent a viable alternative to rescue populations and maintain biodiversity and ecosystem functioning.

(3) We will gain insight by using the network approach to study interaction dynamics and the effects of coevolutionary rescue on biodiversity dynamics and ecosystem functioning.

Metacommunity ecology has been most successful at providing a theoretical predictive framework for competitive metacommunities. Studying coevolution in pairwise interactions may function for a subset of organisms (intimate interactions?), but most species are immersed in complex interaction networks. We will advance our knowledge on species interactions by applying the network approach to understand how species may deal with biodiversity changes and how ecosystem functioning will be maintained. By recognizing that we need networks, we set the scene to consider rewiring (species turnover and switchts in interaction partners) as another mechanism to rescue species from mutualistic partners and/or food loss.

(4) We need to understand the role of species dispersal in local and regional biodiversity dynamics and ecosystem functioning because incorporating species movement (i.e. gene flow) may change predictions based on closed communities; therefore we need to incorporate the co-evolving metacommunity approach.

Species dispersal is another ecological and/or evolutionary rescue mechanism. Incorporating species movement will broaden our understanding of rescue mechanisms and improve model predictions on biodiversity dynamics and ecosystem functioning in a rapid changing world.

(5) We need a trait-based approach for understanding biodiversity dynamics and ecosystem functioning in a rapid changing world. (Here we can also incorporate the niche perspective that Ruben mentioned and draw during the workshop).

Species traits driving species responses to biotic and abiotic selective pressures as well species movement, and the evolutionary relationships among these traits may determine trait variation and species persistence at different spatial scales. Thus, a coevolutionary multi-trait-based approach will be useful for developing the concept of functional persistence and predicting biodiversity dynamics and ecosystem functioning in a rapid changing world.

2. Understanding biodiversity and ecosystem dynamics in a rapid changing world: Evolutionary, coevolutionary and ecological rescue.

(We have a general introduction discussing five items outlining the main gaps in which insights from coevolutionary metaecosystems are missing. We are entering now in a section in which we need to give insights from the ecological, evolutionary and coevolutionary perspectives. This section will focus on processes favouring species rescue, i.e. processes allowing species

persistence in the face of continuous biotic and abiotic stressors. We use the idea of rescue because we think it encloses ecological, evolutionary and coevolutionary processes at different levels of biological organisation -populations, communities and metacommunities.

Rescue and therefore persistence can be achieved by different mechanisms, such as dispersal, interaction rewiring, evolution and coevolution (see Bell 2017, Nuismer et al. 2018). Section 2 aims to describe all these mechanisms in order to introduce the need of the BAD COMETA approach to study biodiversity dynamics and ecosystem functioning in a rapid changing world.

Below, we approach the main points from the (co)evolutionary and ecological side, and we want you to contribute with your work and background. We put in blue the name of people that may contribute with that part of the article).

2.1. What is rescue? (Cecilia and Julia) Processes allowing species persistence in the face of continuous biotic or abiotic stressors. Ecological, evolutionary and coevolutionary rescue. Main references: Bell 2017 ARES, Nuismer et al. 2018.

2.2. Ecological rescue (In this section we would like to introduce the importance of considering rewiring -and therefore another advantage of studying ecological networks- and dispersal -the metacommunity approach- as rescue mechanisms to predict biodiversity dynamics and ecosystem functioning in a rapid changing world. Dispersal can also be considered as an evolutionary mechanism of rescue -see Bell 2017- but we will keep it here and mention this at the end of the section)

2.2.1. Rewiring (key concepts: species turnover, interaction turnover, rewiring interaction partners as a stabilizing process in communities)

What is rewiring?

Why rewiring can be considered as an ecological mechanism promoting species persistence in a rapid changing world. (Catherine, Malena and Ruben, it would be great if you can contribute to this section by thinking on your study systems. For instance, Catherine, how preference patterns from hummingbirds influence rewiring and therefore ecological rescue? If flower morphotypes are lost (reducing trait variance), how this may affect species persistence? Do hummingbirds switch to expand their niche width or do they explore new plant populations across new habitats? Is there any insight from how coevolution might alter ecological rescue? Malena and Ruben, those questions may also apply to your systems. Malena, it would be great to think here about the persistence of interactions and why some are rewired while others are more resistant to change and others are lost. Also Malena and Luisjo, could you develop some insights about temporal and spatial dynamics of species and interactions diversity (metanetworks) in anthropogenic (rapid changing) landscapes? A more challenging question here would be how central or peripheral populations are differently affected by rewiring. Do you have any insight related to this point? Finally, if you think in coevolution, how may it change species persistence in a rapid changing world?)

Mathematical models predictions (mostly focused on trait-based models) Catherine, Luisjo, it would be great if you could provide some insight here about existing models considering rewiring (either in local networks --most of the existing models/empirical patterns-- or metanetworks).

Empirical evidence Catherine, Malena and Ruben, can you provide some empirical patterns from your own data or data from other sources?

2.2.2. Dispersal (François, Matías, Julia) (This section should introduce the importance of dispersal as an ecological rescue mechanism in the context of the metacommunity/metaecosystem approach to study biodiversity dynamics and ecosystem functioning in a rapid changing world. Key concepts: dispersal, metacommunities/metaecosystems, network approach to metacommunities --interactions and space--, evolving metacommunities)

(Julia) Why dispersal can be considered an ecological mechanism promoting species persistence in a rapid changing world: the importance of looking at communities as *open communities* (metacommunities) to understand and predict biodiversity dynamics in changing landscapes. Developing the concept of source-sink dynamics to introduce the idea of why we need to understand the joint effects of environmental heterogeneity and dispersal on biodiversity dynamics.

Predictions of mathematical models (This section should summarize metacommunity models dealing with the importance of dispersal to investigate biodiversity dynamics). Which metacommunity models should we include here? Those that already integrate the network approach to interactions? (i.e. not just competitive metacommunities?). If we incorporate the network approach here, then section 2.2.3. can be about evolving competitive metacommunities that include the network approach. (François, Julia, Matías)ok.

Species-centered models. Models considering dispersal as a species trait (i.e. each species has its own dispersal ability, associated or not with other species traits). Models dealing with species dispersal in competing metacommunities. Models dealing with trophic or mutualistic interactions and dispersal. Trait-based metacommunity models. (François, can you provide insight in this section?) (Chaianunporn and Hovestadt 2012) Travis et al. (2013) (Doebeli 1995; Doebeli and Ruxton 1997) Finally, (Pillai et al. 2012) (Ronce et al. 2000; Jansen and Vitalis 2007)

Patch-centered models. Models considering dispersal as a landscape feature (migration of a given number of individuals among connected patches). Effects of patch connectivity/centrality on biodiversity. Effects of landscape configuration and species dispersal on biodiversity; models suitable-unsuitable habitat. (Matías, can you provide insight in this section?) OK

Models considering dispersal and environmental heterogeneity. (Julia)

Empirical evidence (Julia, Matías)

2.3. Evolutionary and coevolutionary rescue. (In this section we would like to introduce the importance of considering evolutionary and coevolutionary mechanisms to predict species rescue in the context of individuals dealing with (1) changes in interspecific interaction partners (without considering metapopulations -dispersal- or multiple species -communities-), (2) environmental changes (changes in abiotic conditions), (3) cascade effects (o i.e. considering closed communities and interaction networks), and therefore to understand biodiversity dynamics and ecosystems functioning in a rapid changing world).

2.3.1. Biotic traits and species interactions

Rapid evolution, eco-evolutionary feedbacks (Philine, Cas, and Lutz, it would be great if here you can provide main patterns coming from experimental evolution and coevolution. Can you speculate a bit about the role of coevolution as a rescue mechanism? For example, as a force promoting the emergence of new phenotypes and changes in phenotype frequency with time, i.e. increasing trait variation).

Mathematical models predictions (key concepts: red queen, adaptive foraging, adaptive networks, feedbacks) (Weini and Ayana it would be great to have your insight here. Even if most models of coevolution do not take space into account, or do not study specifically rescue, we see a

potential here to discuss about how coevolutionary dynamics can increase or decrease species persistence and therefore drive biodiversity dynamics and ecosystem functioning.

All organisms experience intra- and inter- specific interactions with other individuals sharing the same environment such as resource and space. These interactions introduce selection pressure on different traits of an organism. For example, a prey may need to allocate its limited energy to reproduction as well as defense against predators. A host may have a trade-off between its growth and evolution of resistance to pathogens. The shape of trade-off curves are important to determine the diversity level in populations and ecological communities. A weak (cheap) trade-off between two traits, where the improvement of one trait is associated with a small decrease in the other trait, allows a coexistence of intermediate phenotypes in the same species. On the contrary, a strong (costly) trade-off often leads to extreme phenotypes, where individuals invest in one of the two traits (Jones et al., JMB, 2007; Jones et al., PRSB, 2009, Beardmore et al., Nature, 2011, Kasada et al., PNAS, 2014). Recently, by assuming trade-offs between fecundity and competitive abilities in computational simulations of evolving communities, Farahpour et al. showed that the highest species diversity is achieved under intermediate (between weak and strong) shapes of trade-offs (Farahpour et al., Elife, 2018). The fact that the shape of trade-offs can also evolve, brings another level of complexity between intra- and inter- specific interactions and the diversity level in biological systems. Huang et al. has shown that the trade-offs of prey growth and defense can change from a weak to a strong correlated shape

Weini, below we discuss about “Abiotic traits and environmental changes” (subsection 2.3.2.) with a section about Niche construction -- some of your approaches about public goods games and abiotic-biotic trade-offs can be discussed here -- but take it easy, if you feel it is too much speculation, then just skip it. Ayana, we are not considering coevolutionary diversification dynamics (cdd) models, but it would be great to think in a paragraph or two about how cdd could affect rescue, not only of one species or phenotype, but entire groups or communities and at a much broader spatiotemporal scales, in the sense of rescuing ecosystem functions.)

Empirical evidence (Lutz, here the opportunity to provide feedback about the outcomes of your experiment in an explicit landscape? Do you have insights connecting coevolution and the idea of coevolutionary rescue?)

2.3.2. Abiotic traits and environmental changes

(key concepts: phenotypic plasticity, rapid evolution, local adaptation, eco-evolutionary feedbacks, niche construction) (Blake, here it would be nice if you can provide insights about the effects of abiotic factors on co-evolutionary rescue. This can be pretty broad and rather speculative. You can think about feedbacks between traits related to environment (abiotic traits) and species interactions, or does it make sense to think about feedbacks between abiotic traits and coevolution? How does niche construction can alter coevolutionary rescue? Does it help to rescue populations?).

Mathematical models predictions? Who can contribute here?

Empirical evidence (Philine, Lutz, Cas, could you give us some insights about environmental stress affecting coevolutionary dynamics?)

2.3.3. Evolution and coevolution in complex communities (Cecilia will introduce the approaches for modeling coevolution in networks)

- (i) Concepts and hypotheses
- (ii) Mathematical models predictions

2.4. Eco-evolutionary rescue

2.4.1. Evolving metacommunities (Cecilia and Julia) (Here we will introduce the previous approaches developed for studying eco-evolutionary rescue in competitive metacommunities and food webs. This has been called evolutionary metacommunities and we will discuss how the predictions --monopolization, etc-- can change when considering coevolution)

3. Moving towards a multi-trait-based approach for coevolutionary interactions in metaecosystems (Carlos, Cecilia, Julia)

In this section we will connect multiple traits evolution and biodiversity dynamics to understand how trait architecture, migration, and coevolutionary selection differentially shape rescue dynamics in the face of different anthropogenic threats to biodiversity. Such understanding will benefit from: (1) investigating how the relationship among biotic, abiotic and dispersal traits determines species responses to the consequences of anthropogenic threats (i.e. reduction in intraspecific variance, conspecific and interaction partners loss, local extinctions, habitat destruction); (2) constructing and evaluating multi-trait-based models considering coevolutionary dynamics in species-rich interaction networks in metacommunities, which may allow making predictions about biodiversity dynamics under multiple threats.

3.1. The multi-trait-based approach for coevolutionary interactions (Julia)

3.1.1. Why a trait-based approach? General advantages of adopting a trait-based approach. Why we should care about relationships among the *BAD* traits (their co-variation may matter for species persistence). How correlations (“the magic trait”) or no correlation (modular structure) among traits may impact species persistence?

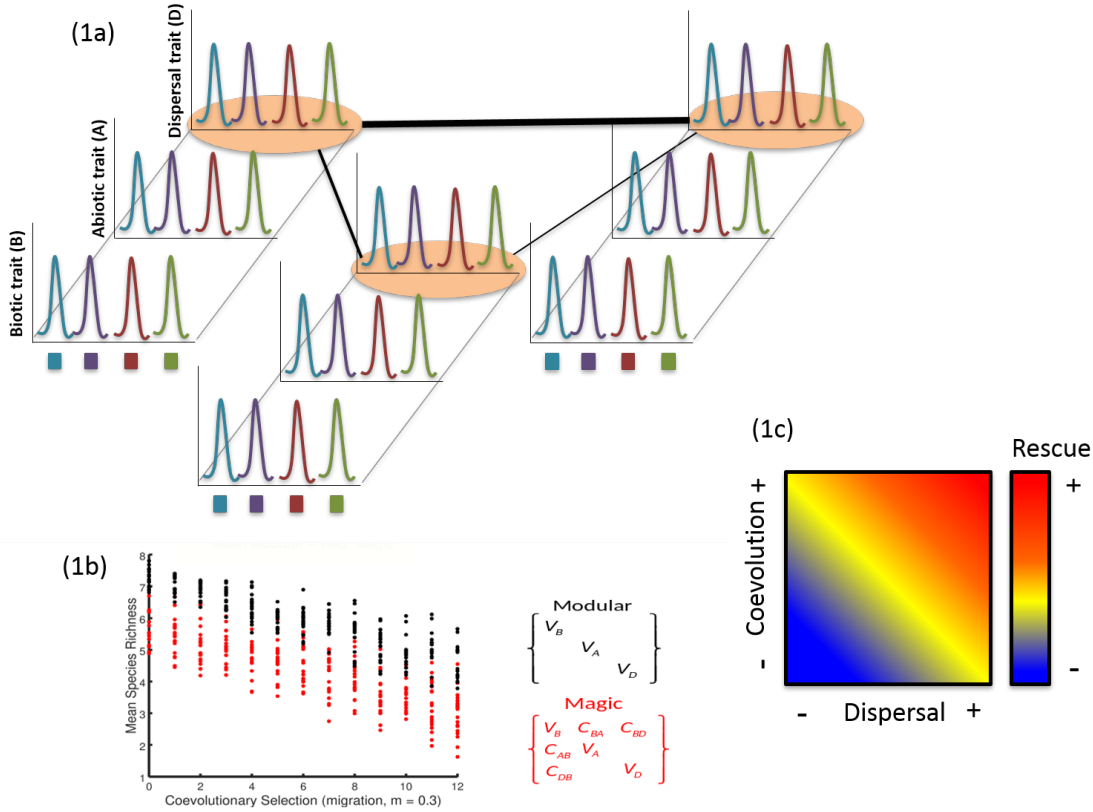


Figure 1 a) Cartoon of the BAD multitrait biodiversity model: 1a) Four species, blue, violet, red and green color, are shown in each site. Each species has three traits, the biotic, the abiotic and the dispersal trait. The dispersal trait connects sites throughout migration (pink). 1b) Mean species richness decays with the strength of coevolutionary selection for the biotic trait in two scenarios, the modular (black) and the magic (red) scenario. Adding interactions among traits, the magic scenario, decreases the number of species for all the gradient of coevolutionary selection, compared to the modular scenario. 1c) Heatmap with rescue hotspots and coldspots

Box 1. A coevolutionary IBM to study biodiversity dynamics and ecosystem functioning in rapid changing landscapes: The BAD multitrait biodiversity model

We outline a model to evaluate the effect of trait architecture (i.e., the correlations among biotic, abiotic and dispersal traits) and rescue processes on biodiversity dynamics. We implement a spatially explicit multitrait-based model to explore how biotic, abiotic and dispersal traits architecture together with coevolutionary selection and dispersal affect biodiversity dynamics and rescue (Figure 1a). We explore two trait architectures rooted in empirical and theoretical studies (refs): the modular and the magic trait architectures. The modular architecture takes into account independent traits, i.e. each trait contributes independently to fitness (Figure 1b: Modular). The magic trait architecture incorporates the biotic trait as the main contributor to fitness with the abiotic and the dispersal traits depending on the evolution of the biotic trait (Fig 1b: Magic).

Our model considers a landscape with many habitat patches inhabited by resource and consumer populations comprising several individuals each, which biotic, abiotic and dispersal trait values are sampled from three normal distributions (biotic, z_b , abiotic, z_a , and dispersal, z_d , represented as $z = [z_b, z_a, z_d]$ for phenotype of individual i). Populations are located in a network of discrete habitat patches connected by migration events. The local population demography is driven by the temporal dependent fitness function accounting for trait architecture following

$$W(z)_x^t = \exp [-\gamma (([z_x^t - \theta_x]^2)^T \omega^{-1} [z_x^t - \theta_x]^2)] \quad (1)$$

where z_x^t is the vector of trait values of phenotype i at time t and site x , θ_x is the multivariate fitness optimum, ω is the covariance matrix describing trait architecture (Figure 1b) (Lande, 1980, Melo & Marroig, 2014) (Melo and Marroig 2015), and γ determines the interaction sensitivity to deviations from the biotic (i.e., coevolutionary selection strength), abiotic and dispersal optimum. Demographic and trait evolutionary dynamics depends on the scenario of trait architecture being considered. If we consider the modular scenario (Figure 1b), i.e. the covariance matrix, ω , is diagonal, then there is no correlated stabilizing selection. In the magic trait scenario, the covariance matrix is not diagonal and there is correlated stabilizing selection with the biotic trait determining the selection on the other traits. The biotic optimum changes in time. We define it as the minimum and maximum mean trait distance across all populations in site x for the consumer and the resource biotic trait value, respectively. The dispersal optimum is given by the mean among-patch distance in the landscape (i.e., each site has coordinates from which we calculate the Euclidean distance between each pair of sites). Phenotypes with dispersal trait value near to such optimum maximize fitness for this trait only in the modular but not in the magic trait scenario. The abiotic optimum is taken initially as the mean of the distribution and it remains the same during the temporal dynamics.

The total number of individuals is constant (zero sum model) and at each time step a certain percentage of individuals are selected randomly to die and are replaced by birth or migration. The time dependent fitness function in site x , $W(z)_x^t$, determines the mother of the offspring. To obtain the phenotypic value for the offspring, $z_o = [z_{ob}, z_{oa}, z_{od}]$, we add to each trait value of the mother the environmental deviation, e , taken from a multivariate uniform distribution with range 0.001 in all dimensions. We run our model G generations each following the selection and dispersion scenarios from the modular or magic scenarios to produce a new generation per site. To evaluate the importance of coevolution and dispersal as rescue mechanisms in modulating biodiversity dynamics, we explore how the formation of hot and cold spots of species persistence in local and regional biodiversity changes in a gradient of dispersal and coevolutionary selection strength. Mean species richness decays with the strength of coevolutionary selection for the biotic trait in the two scenarios, the modular (black) and the magic (red) scenario (Figure 1b). Adding interactions among traits, the magic scenario, decreases the number of species for all the gradient of coevolutionary selection, compared to the modular scenario. (Q3: mention the initial values: number of species, population size, initial trait variance and covariance values, and the number of generations per replicate. Codes and outputs can be downloaded from <https://github.com/melian009/Cometa>).

3.2. An empiricist guide to test predictions of coevolutionary metaecosystems models (Ruben, Catherine, Malena, Matías) OK

(Here we would like to discuss conceptual relationships among *BAD* traits that have been proposed and empirical patterns reported. The candidate *BAD* traits that should be measured by field ecologists/evolutionary ecologists, ideally for different interaction types and organisms, including the four datasets we have summarized during the workshop. We will consider that you already got some traits that measure species matching, as we discussed during the workshop, and guess that you will be interested in discussing how to measure species dispersal/movement ability --we started some discussions at the workshop-- and species responses to environment, i.e. which trait may determine species distributions associated to environmental variables? For example, seed size in plants, body size in bees, etc. For dispersal, we can discuss dispersal syndromes for plants, intertegular distance for bees, wing size for birds).

3.2.1. Biotic traits (Ruben, Catherine, Malena, Matías) OK

- Which empirical traits are promising traits to estimate species interaction strength and thus the strength of reciprocal selection in your study system?
- Can we speculate about the functional mechanism linking traits to fitness (trait *matching/mismatching/barrier*) in your study system?
- Can we estimate the strength of species interaction selection based on the distribution trait complementarity and convergence in the metacommunity?

3.2.2. Abiotic traits (Ruben, Catherine, Malena, Matías)

- Which empirical traits are promising traits to estimate species response to environmental filters?
- Can we estimate the strength of environmental filter selection based on species traits distribution in the metacommunity?

3.2.3. Dispersal traits (Ruben, Catherine, Malena, Matías)

- Which empirical trait are promising traits to estimate species dispersal ability (i.e. the ability to move among patches/communities)?
- Can we estimate the strength of selection on dispersal ability based on species traits distribution in the metacommunity? OK

3.2.4. Inferring empirical correlations among *BAD* traits (Cecilia, Julia, Carlos)- Is it possible to infer the architecture of *BAD* traits? How can we empirically build the matrix of *BAD* traits architecture? Approaches to study quantitative traits integration. See Armbruster et al. 2014.

3.3. Linking model predictions with data (Carlos, Matias?) OK

Which methods can be used to compare COMETA model predictions with empirical data?

- ABC, Bayesian methods
- CATs analysis to understand the importance of neutral and niche effects on trait distribution.

4. Concluding remarks and future directions

References

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